Higher bioaccessibility of iron and zinc from food grains in the presence of garlic and onion.

Bioavailability of micronutrients iron and zinc is particularly low from plant foods. Hence there is a need to evolve a food-based strategy to improve the same to combat widespread deficiencies of these minerals in a population dependent on plant foods. Dietary sulfur-containing amino acids have been reported to improve the mineral status of experimental animals. Our objective was to examine whether sulfur compound-rich Allium spices have a similar potential of beneficially modulating the mineral bioavailability. In this context, we examined the influence of exogenously added garlic and onion on the bioaccessibility of iron and zinc from food grains. Two representative cereals and pulses each were studied in both raw and cooked condition employing two levels of garlic (0.25 and 0.5 g/10 g of grain) and onion (1.5 and 3 g/10 g of grain). The enhancing effect of these two spices on iron bioaccessibility was generally evidenced in the case of both the cereals (9.4-65.9% increase) and pulses (9.9-73.3% increase) in both raw and cooked conditions. The two spices similarly enhanced the bioaccessibility of zinc from the food grains, the extent of increase in cereals ranging from 10.4% to 159.4% and in pulses from 9.8% to 49.8%. Thus, both garlic and onion were evidenced here to have a promoting influence on the bioaccessibility of iron and zinc from food grains. This novel information has the potential application in evolving a food-based strategy to improve the bioavailability of trace minerals and hence contributes to the human health benefit.